WORKED SOLUTION

STEP I 2007, Question 7

Pure • Distance Between Lines

STEP I 2007, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q7

The key idea

Write down a general point on each line, subtract, and square the length: D^2 is then a quadratic in the two parameters. The trick is to write that quadratic as a sum of squares, because a sum of squares is smallest when each square is as small as it can be. In part (ii) the same method works, and the way the squares fall apart for different values of k tells us the geometry.

Solution

Part (i)

A general point of L_1 is $P = (1+2\lambda,\ 2\lambda,\ 2-3\lambda)$ and a general point of L_2 is $Q = (4+\mu,\ -2+2\mu,\ 9-2\mu)$. So

$$\overrightarrow{PQ} = \begin{pmatrix} 3+\mu-2\lambda \\ -2+2\mu-2\lambda \\ 7-2\mu+3\lambda \end{pmatrix}$$

and

$$D^2 = (3+\mu-2\lambda)^2 + (-2+2\mu-2\lambda)^2 + (7-2\mu+3\lambda)^2.$$

Collecting terms coefficient by coefficient:

$$\mu^2: \quad 1+4+4=9, \qquad \lambda^2: \quad 4+4+9=17, \qquad \lambda\mu: \quad -4-8-12=-24,$$
$$\mu: \quad 6-8-28=-30, \qquad \lambda: \quad -12+8+42=38, \qquad \text{const}: \quad 9+4+49=62,$$

so $D^2 = 9\mu^2 + 17\lambda^2 - 24\lambda\mu - 30\mu + 38\lambda + 62$.

Now expand the proposed form:

$$\begin{aligned}(3\mu-4\lambda-5)^2 + (\lambda-1)^2 + 36 &= (9\mu^2+16\lambda^2+25-24\lambda\mu-30\mu+40\lambda) \\ &\quad + (\lambda^2-2\lambda+1) + 36 \\ &= 9\mu^2+17\lambda^2-24\lambda\mu-30\mu+38\lambda+62.\end{aligned}$$

The two agree, so

$$D^2 = (3\mu - 4\lambda - 5)^2 + (\lambda - 1)^2 + 36,$$

as required.

Ask yourself Why can we make both squares zero at once?

Each square is at least 0, so $D^2 \geq 36$ for every choice of λ and μ. Equality needs $\lambda = 1$ and $3\mu = 4\lambda + 5 = 9$, that is $\mu = 3$. These two conditions are compatible (the first fixes λ, and then the second fixes μ), so $D^2 = 36$ really is attained. Hence the minimum distance between the lines is

$$D_{\min} = 6,$$

attained at

$$\lambda = 1 : \ P = (3,\, 2,\, -1) \quad \text{on } L_1, \qquad \mu = 3 : \ Q = (7,\, 4,\, 3) \quad \text{on } L_2.$$

As a check, $\overrightarrow{PQ} = (4, 2, 4)$ has length $\sqrt{16 + 4 + 16} = 6$, and it is perpendicular to both direction vectors: $(4, 2, 4) \cdot (2, 2, -3) = 8 + 4 - 12 = 0$ and $(4, 2, 4) \cdot (1, 2, -2) = 4 + 4 - 8 = 0$. That is exactly what we expect of the shortest segment joining two lines.

Part (ii)

A general point of L_3 is $P = (2,\ 3 + \alpha,\ 5)$ and a general point of L_4 is $Q = (3 + 4k\beta,\ 3 + (1 - k)\beta,\ -2 - 3k\beta)$. So

$$\overrightarrow{PQ} = \begin{pmatrix} 1 + 4k\beta \\ (1 - k)\beta - \alpha \\ -7 - 3k\beta \end{pmatrix}$$

and

$$\begin{aligned} D^2 &= (1 + 4k\beta)^2 + ((1 - k)\beta - \alpha)^2 + (7 + 3k\beta)^2 \\ &= ((1 - k)\beta - \alpha)^2 + 25k^2\beta^2 + 50k\beta + 50 \\ &= ((1 - k)\beta - \alpha)^2 + 25(k\beta + 1)^2 + 25, \end{aligned}$$

since $(1 + 4k\beta)^2 + (7 + 3k\beta)^2 = 1 + 8k\beta + 16k^2\beta^2 + 49 + 42k\beta + 9k^2\beta^2$.

Whatever β is, we can choose $\alpha = (1 - k)\beta$ to make the first square zero, because α appears nowhere else. So the question is how small $25(k\beta + 1)^2$ can be made, and this is where the two cases separate.

Ask yourself What changes when $k = 0$?

If $k \neq 0$ we can take $\beta = -1/k$, which makes $k\beta + 1 = 0$. If $k = 0$ the term $25(k\beta + 1)^2$ is stuck at 25 whatever β is.

Case $k \neq 0$**.** Every square is non-negative, so $D^2 \geq 25$, with equality at $\beta = -1/k$, $\alpha = (1 - k)\beta = (k - 1)/k$. The minimum distance is

$$D_{\min} = 5,$$

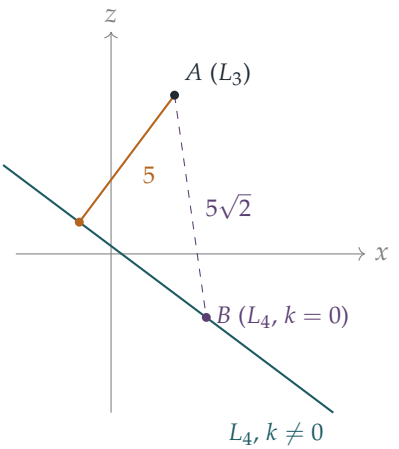
whatever the (non-zero) value of k. The closest points are $P = (2,\ 3+\frac{k-1}{k},\ 5)$ and $Q = (-1,\ 3+\frac{k-1}{k},\ 1)$, with $\overrightarrow{PQ} = (-3,0,-4)$ of length 5. Since $D_{\min} > 0$ the lines never meet, and their directions $(0,1,0)$ and $(4k, 1-k, -3k)$ are not parallel when $k \neq 0$ (the first and third components of the second are not both zero). So the lines are *skew*, and 5 is the length of their common perpendicular, which is unique.

Case $k = 0$. Now L_4 has direction $(0,1,0)$, the same as L_3: the lines are *parallel*. Here $D^2 = (\beta - \alpha)^2 + 50$, minimised whenever $\alpha = \beta$, so

$$D_{\min} = \sqrt{50} = 5\sqrt{2},$$

and it is attained by infinitely many pairs of points: any point of one line together with the point of the other directly opposite it (same y-coordinate). This is just the constant distance between two parallel lines.

The geometry. Both lines are easiest to see by looking straight down the y-axis, that is, projecting onto the xz-plane. Since L_3 runs parallel to the y-axis, it projects to the single point $A = (2,5)$, and distances perpendicular to L_3 are unchanged by the projection. For $k \neq 0$, L_4 projects to the line through $(3,-2)$ with direction $(4,-3)$ (the $1-k$ component disappears), and the distance from A to that line is $|1\cdot(-3) - (-7)\cdot 4|/5 = 5$, independent of k. For $k = 0$, L_4 projects to the single point $B = (3,-2)$, and the distance is $AB = \sqrt{1+49} = 5\sqrt{2}$.



So the answer is 5 when $k \neq 0$ (skew lines, one shortest segment, which is perpendicular to both) and $5\sqrt{2}$ when $k = 0$ (parallel lines, the distance is the same all along).

Checks

In part (i), substituting $\lambda = 1$, $\mu = 3$ gives points 6 apart, and $\overrightarrow{PQ}$ is perpendicular to both lines. A numerical search over a grid of (λ, μ) found nothing below $D^2 = 36$. In part (ii), minimising D^2 numerically for $k = 0.001,\ 0.3,\ 1,\ -2,\ 5$ gave 25 each time, and 50 for $k = 0$. Notice that $D_{\min}$ jumps from 5 to $5\sqrt{2}$ at $k = 0$. For tiny k the closest points are a long way out, near $\beta = -1/k$, so the limit doesn't have to be continuous.

Common slips

- Forgetting to check that both squares can vanish *simultaneously*. Here they can, but that has to be said, not assumed.
- Dividing by k when $k = 0$: the choice $\beta = -1/k$ is only available for $k \neq 0$, which is exactly why the two cases arise.
- Calling the $k \neq 0$ lines "intersecting". They are not parallel, but $D_{\min} = 5 > 0$, so they are skew.

Beyond STEP

Completing the square in several variables is the start of the theory of quadratic forms: writing D^2 as a sum of squares is diagonalising a symmetric matrix, and the minimum is a least-squares problem. The perpendicularity we checked at the end is the "normal equations" of least squares, and the general formula $|(\mathbf{b}-\mathbf{a})\cdot(\mathbf{u}\times\mathbf{v})|/|\mathbf{u}\times\mathbf{v}|$ for the distance between skew lines comes from the same geometry. Strang, *Introduction to Linear Algebra*, covers projections and least squares beautifully.

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